

Best Exercises for Osteoarthritis Knee Pain at Home

Practical NHS-backed advice on best exercises for osteoarthritis knee pain at home.

By **Living With Arthritis UK** · Reviewed by **Louis Maxwell** (HCPC Registered Physiotherapist) · 2026-07-24 · livingwitharthritis.org.uk

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Quick Answer: The most effective home exercises for knee osteoarthritis are quad sets, straight leg raises, sit-to-stand practice, mini squats, and calf raises. These strengthen muscles around the knee, reducing load on the joint. NICE recommends exercise as a core treatment — 20–30 minutes daily produces measurable pain reduction.

Exercise is one of the most effective treatments for knee OA — as effective as anti-inflammatory medication for many people, with no side effects. The key is building strength in muscles that support the knee.

The 5 Best Home Exercises

Strengthening Exercises

- **Quad sets:** Sit with leg straight, tighten thigh muscle pressing back of knee toward floor, hold 5 seconds. 15 reps, twice daily
- **Straight leg raise:** Lie on back, tighten thigh, raise straight leg to height of bent knee, hold 3 seconds. 12 reps each leg
- **Sit-to-stand:** From firm chair, lean slightly forward, push through heels to stand, lower slowly. 10 reps, 2–3 times daily

Low-Impact Strengthening

- Mini squats: Stand holding worktop, bend knees to 20–30 degrees, hold 3 seconds, straighten. 12 reps
 - Calf raises: Stand holding a surface, rise onto tiptoes slowly, lower slowly. 15 reps
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Building Up Safely



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Start with one set daily for the first week. Build to two sets in week two, three sets by week three. Most people notice pain improvement after 4–6 weeks. Walking on flat surfaces (starting 10 minutes) is also an excellent complement to home exercises.

Q&A;

Question: Will exercise make my knee OA worse? **Short answer:** No — gentle strengthening exercise does not worsen OA and is strongly recommended by NICE. Complete rest actually increases joint load and worsens OA over time.

Question: Should I exercise when my knee is swollen? **Short answer:** During active swelling, reduce intensity and stick to gentle range-of-movement exercises only. Once swelling settles, resume strengthening exercises gradually.

Question: How long until I see improvement? **Short answer:** Most people notice reduced pain and improved function after 4–8 weeks of daily exercise. Strength gains continue for 3–6 months with consistent practice.

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